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Software Engineering Lab 1

Requirements Engineering & UML Use-Case Modelling

Problem Statement #37 – Group Travel Expense Simplification Engine

Actors: Group Traveler, Trip Admin

1. Complete Requirements Table

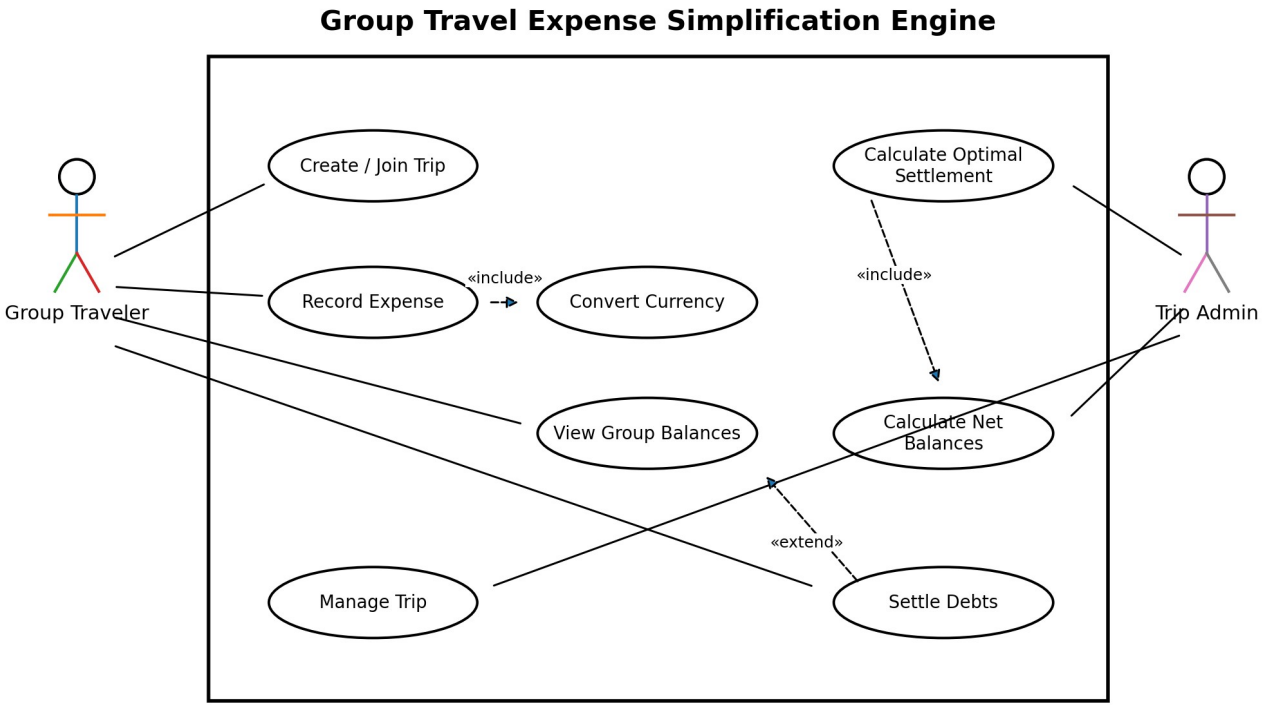
ID	Type	Priority	Description	Acceptance Criteria	Rationale
FR-001	Functional	High	The system shall record group expenses with payer, amount, currency, date, participants, and split details.	Pass: Expense is saved with all required fields and appears in the trip ledger. Fail: Any required field is missing or the expense is not stored.	Provides the core data needed to calculate each traveler's share and final settlement.
FR-002	Functional	High	The system shall convert foreign-currency expenses into the trip base currency using the applicable daily exchange rate.	Pass: Converted amount matches the stored daily rate for the expense date. Fail: The system uses an unavailable or incorrect rate or produces an inconsistent converted value.	Ensures expenses in different currencies can be compared and settled in one common currency.
FR-003	Functional	High	The system shall calculate each traveler's net balance from all recorded expenses and shares.	Pass: The sum of all positive balances equals the absolute sum of all negative balances. Fail: Group balances do not reconcile to zero.	Net balances are the input required for efficient debt settlement.
FR-004	Functional	High	The system shall generate an optimal debt settlement graph that minimizes the number of peer-to-peer payment transactions.	Pass: All traveler balances reach zero after proposed payments and the settlement uses the minimum feasible payment edges. Fail: Balances do not reconcile or the result uses unnecessary payment edges.	Directly implements the main purpose of the engine: simplifying group settlement.
FR-005	Functional	Medium	The system shall allow travelers and the trip admin to view the expense ledger, balances, and recommended settlement transactions.	Pass: Authorized users can view current expenses, balances, and settlement instructions. Fail: Data is missing, stale, or visible to an unauthorized user.	Makes the calculated result understandable and actionable for the group.

2. Non-Functional Requirements

ID	Type	Priority	Description	Acceptance Criteria	Rationale
NFR-001	Performance & Security	High	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses, while protecting trip data from unauthorized access.	Pass: Benchmarking confirms <100 ms at the stated load and access-control tests prevent unauthorized trip-data access. Fail: Either target is violated.	The assignment requires fast settlement for realistic group sizes and secure handling of financial information.
NFR-002	Reliability & Data Integrity	High	The system shall preserve expense, exchange-rate, balance, and settlement data consistently and shall prevent a failed update from leaving a partially applied transaction.	Pass: Tests show no lost/duplicated expenses and failed operations roll back without corrupting balances. Fail: Any inconsistent or partial state is observed.	Accurate financial records are essential because even a small data error can produce incorrect settlement amounts.

3. UML Use-Case Diagram

Actors: Group Traveler and Trip Admin. The model contains the primary use cases and explicitly includes both «include» and «extend» relationships.



Dashed arrow = «include» / «extend»; solid line = actor association.

4. Use-Case Flow Specification

Use Case: Compute Optimal Debt Settlement

Preconditions: The trip exists; participating travelers are registered; expenses have been recorded; each expense has a valid currency and applicable exchange rate; and the user is authorized to access the trip.

Postconditions: Each traveler has a reconciled net balance; the settlement graph contains payment instructions that bring all balances to zero; and the recommended transactions are available for the group to view.

Main Success Scenario: 1. User opens the trip settlement screen.
2. System retrieves all recorded expenses and applicable exchange rates.
3. System converts foreign-currency expenses to the trip base currency.
4. System calculates each traveler’s net balance.
5. System separates travelers into creditors and debtors.
6. System matches debtors to creditors and computes payment amounts using a minimum-transaction settlement strategy.
7. System validates that the proposed payments settle all balances to net zero.
8. System displays the recommended peer-to-peer payments.

Alternate Flow: A1. If an exchange rate required for an expense is unavailable or invalid, the system flags that expense, does not use an unreliable conversion, and asks the user/admin to provide or correct the rate. Settlement calculation is paused until the data is valid.